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Week 1 Module

C and C++ programming

(1) C Syntax and Basic Structure

C is a procedured and Structured programming language. Every standard C program has a precise, structured layout.

Header files (#include <stdio.h>)

This pre-processor directive includes the Standard Input/output Library. It allows the program to use the core functions, such as printf() for output and scanf() for input.

Main function (int main())

This is the mandatory entry point of any C program. Execution starts at the opening brace of main(). Execution ends at the terminating semicolon of return 0;

output function (printf)

Unlike C++, C uses `printf()` function to display text on the screen. Double quotes let you pass format specifiers or escape sequences like `\n` (newline).

Semicolon (;): Every statement in C must end with a semicolon. It is a statement terminator for the compiler.

② Conditions and loops

Conditions statements (if-else): used to make decisions. The if block runs when the condition is true. If it is false the program skips it or executes the optional block.

loops (for loop): To repeat a block of code a certain number of times. It is composed of initialization, condition checking and increment/decrement expressions.

Nested loops for star patterns to print two dimensional grid patterns we need a loop in the loop
outer loop :- Manages the rows (vertical program)

Inner loop :- Handles the Columns (prints spaces or stars horizontally for the current flow)

③ programming Modules problems:-

Solve the given star pattern questions

① Square

Q/p :-

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

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Run

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int n = 5; // total rows and columns
6     //outer loop for rows
7     for (int i = 1; i<=n; i++){
8         //inner loop for columns
9         for (int j=1; j<=n; j++){
10             printf("*");
11         }
12         printf("\n"); // move to the next line
13     }
14     return 0;
15 }
```

Output

```
*****
*****
*****
*****
*****

=== Code Execution Successful ===
```


#include <stdio.h>

int main () {

int n=5; // total rows and columns

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

printf ("*");

}

printf ("\\n");

}

return 0;

}

(2) Hollow Square pattern

#include <stdio.h>

int main () {

int n=5;

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

if (i==1 || i==n || j==1 || j==n) {

printf ("*");

} else {

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Run

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```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int n=5;
6     for(int i=1; i <=n; i++) {
7         for(int j=1; j<=n; j++){
8             // print star only boundaries
9             if (i==1 || i==n || j==1 || j==n){
10                 printf("*");
11             }else{
12                 printf(" ");// print empty space
13             }
14         }
15         printf("\n");
16     }
17     return 0;
18 }
```

Output

Clear

* *
* *
* *

=== Code Execution Successful ===

JS
TS

printf (" ");

}}

printf ("*\n");

} return 0;

}

O/p :-

* * * * *

*

*

*

*

*

*

*

*

*

*

*

③ Triangle

#include <stdio.h>

int main () {

int n = 5;

for (int i = 1; i <= n; i++) {

for (int j = 1; j <= i; j++) {

printf ("*");

} printf ("\n");

} return 0;

}

O/p :-

*

* *

* * *

* * * *

* * * * *

#ThereIsOnlyOne

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Run

Output

Clear

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3 int main() {
4     int n = 5;
5     for(int i= 1; i<= n; i++) {
6         //inner loop runs up to the currents rows number 'i'
7         for (int j= 1; j<= i; j++){
8             printf("*");
9         }
10        printf("\n");
11    }
12
13    return 0;
14 }
```

```
*
**
***
****
*****

=== Code Execution Successful ===
```

JS

TS

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④ Inverted Triangle

```
#include <stdio.h>
```

```
int main()
```

```
{ int n=5;
```

// outer loop counts down from 'n' to 1

```
for(int i=n; i>=1; i--) {
```

```
for(int j=1; j<=i; j++) {
```

```
printf(" * ");
```

```
} printf("\n");
```

```
} return 0;
```

```
}
```

O/P:-

```
* * * * *
 * * * *
  * * *
   * *
    *
```

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Run

Output

Clear

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int n=5;
6     //outer loop count down from 'n' to 1
7     for (int i=n; i>=1; i--){
8         for (int j=1; j<=i; j++){
9             printf("*");
10        }
11        printf ("\n");
12    }
13    return 0;
14 }
```

```
*****
*****
****
***
**
*

=== Code Execution Successful ===
```

JS

TS

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17-07-2026

⑤ pyramid

```
#include <stdio.h>
int main () {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n + i; j++) {
            printf (" ");
        }
    }
```

// print stars based on odd numbers
formula

```
for (int k = 1; k <= (2 * i - 1); k++) {
    printf ("*");
}
return 0;
}
```

O/p:-

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *

```

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```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main(){
5     int n=5;
6     for(int i=1; i<=n;i++){
7         for(int j=1 ;j<=n-i; j++ ){
8             printf(" ");
9         }
10        //print stars based on odd number formula;2*i-1
11        for(int k=1; k<=(2*i-1); k++){
12            printf("*");
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

Run

Output

```
*
***
*****
*****
*****
```

=== Code Execution Successful ===

23:36:58
17-07-2026

⑥ Inverted pyramid

```
#include <stdio.h>
int main() {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        // print space
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (int k = 1; k <= (2 * i - 1); k++) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

O/P:-

```
* * * * *
* * * *
* * *
* *
*
```

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Run

```
1
2 #include <stdio.h>
3
4 int main(){
5     int n=5;
6     //outer loop handles rows in reverse order
7     for (int i=n; i>=1; i--){
8         for(int j=1; j<=n-i; j++){
9             printf(" ");
10        }
11        for(int k=1; k<=(2*i-1); k++){
12            printf("*");
13        }
14        printf("\n");
15    }
16
17    return 0;
18 }
```

Output

*

=== Code Execution Successful ===

23

ENG

23:45:04

⑦ Half diamond

```
#include <stdio.h>
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf(" * ");
        }
        printf("\n");
    }
    // Lower Half is Spinning inverted Triangle
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            printf(" * ");
        }
        printf("\n");
    }
    return 0;
}
```

O/P :-

```
 *
 * *
 * * *
 * * * *
 * * *
 * *
 * *
```

#ThereIsOnlyOne



main.c



Share

Run

Output

Clear

```
1
2 #include <stdio.h>
3
4 int main(){
5     int n=5;
6     for(int i=1; i<=n; i++){
7         for (int j =1; j<=i; j++){
8             printf("*");
9         }
10        printf("\n");
11    }
12    for(int i=n-1; i>=1; i--){
13        for(int j=1; j<=i; j++){
14            printf("*");
15        }
16        printf("\n");
17    }
18    return 0;
19 }
```

```
*
**
***
****
*****
****
***
**
*

=== Code Execution Successful ===
```

⑧ Diamond:

```
#include <stdio.h>
int main () {
    int n=5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n-i; j++) {
            printf (" ");
        }
        for (int k = 1; k <= (2*i-1); k++)
            printf (" ");
        printf ("\n");
    }
    return
    for (int i = n-1; i >= 1; i--) {
        for (int j = 1; j <= n-i; j++)
            printf (" ");
        for (int k = 1; k <= (2*i-1); k++)
            printf (" ");
        printf ("\n");
    }
    return 0;
}
```

O/p:-

```
      *
    * * *
  * * * *
* * * * *
* * * * *
  * * * *
    * * *
      *
```

#there's Only One

